Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 24)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Explain how occupancy sensors and programmable thermostats reduce energy waste.

[2 marks]

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0 1 . 2 Explain why LED lights use much less electricity than the filament and CFL lamps they replaced, and give **two** other examples of low-energy appliance design.

[3 marks]

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0 1 . 3 Describe **three** changes in human behaviour that reduce domestic energy use.

[3 marks]

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0 1 . 4 Explain why heating water on demand saves energy compared with storing hot water.

[2 marks]

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0 2 A student compared the invertebrates in a stream above and below a sewage outfall using kick sampling.

0 2 . 1 Describe how kick sampling is carried out so that the samples from the two sites are comparable.

[3 marks]

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0 2 . 2 Give **three** limitations of kick sampling.

[3 marks]

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0 2 . 3 Explain how a Surber sampler improves on kick sampling.

[2 marks]

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0 2 . 4 Explain why the invertebrates found give a better indication of long-term sewage pollution than a single measurement of dissolved oxygen.

[2 marks]

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0 3 . 1 Describe how ozone is formed and destroyed naturally in the stratosphere.

[3 marks]

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0 3 . 2 Explain the properties of CFCs that made them useful, and the properties that cause ozone depletion (the Rowland-Molina hypothesis).

[6 marks]

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Useful properties

Properties causing ozone depletion

0 3 . 3 Explain why increased UVB reaching the Earth's surface is harmful to living organisms. Give **three** effects.

[3 marks]

1

2

3

0 3 . 4 State what is meant by an **ozone hole**.

[1 mark]

0 3 . 5 Give **one** reason why concentration in ppm is not used to measure ozone depletion.

[1 mark]

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0 3 . 6 Give **one** reason why the first evidence for ozone depletion came from ground-based measurements.

[1 mark]

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15

0 4 A sand and gravel pit near a town is due to close. Three options for the site are proposed: a wetland nature reserve, housing, or return to farmland.

0 4 . 1 Explain why flooded sand and gravel pits are often developed as wetland nature reserves.

[2 marks]

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0 4 . 2 Explain the conditions that would need to be met for the site to be used for housing or farmland.

[2 marks]

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0 4 . 3 Explain why moving the wildlife from a site before mining begins is rarely completely successful.

[1 mark]

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0 5 . 1 Which of the following is a feature of a river that affects its usefulness as a water source? Tick **one** box.

[1 mark]

A Its total annual discharge ☐

B The albedo of its surface ☐

C Its salinity ☐

D The age of its bedrock ☐

0 5 . 2 Explain why rainwater collection is increasingly important in cities such as those in India.

[2 marks]

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0 5 . 3 Explain why an estuary barrage has fewer land-use conflicts than a reservoir on land but still causes environmental problems.

[2 marks]

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0 6 Table 1 shows four general approaches to pollution control.

0 6 . 1 Complete Table 1 by giving one example of each approach.

[4 marks]

Table 1

Approach	Example
Production prevention	
Prevention of release	
Post-release remediation	
Alternative processes	

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0 6 . 2 Explain why prevention of production is generally preferable to post-release remediation.

[2 marks]

0 6 . 3 Describe how each of the following new technologies helps to control pollution.

[4 marks]

Satellite monitoring of oil spills

GPS ship tracking

Polymer adsorption of heavy metals

Phytoremediation

10

0 7 Table 1 shows the annual emissions of two pollutants in the UK in 1970 and 2015.

Table 1

Pollutant	Emissions in 1970 / thousand tonnes	Emissions in 2015 / thousand tonnes	Main source in 2015
Sulfur dioxide	6 400	220	Energy industries
Nitrogen oxides	2 700	930	Road transport

0 7 . 1 Calculate the percentage reduction in sulfur dioxide emissions between 1970 and 2015.

[2 marks]

Answer ..... %

0 7 . 2 Suggest **three** reasons why sulfur dioxide emissions fell by a much larger percentage than nitrogen oxide emissions.

[3 marks]

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0 7 . 3 Explain why a reduction in NOx emissions would also reduce concentrations of tropospheric ozone and PANs.

[2 marks]

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0 7 . 4 Explain why lichens are used to monitor sulfur dioxide pollution.

[3 marks]

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0 8 The kinetic energy available to a wind turbine is proportional to the cube of the wind velocity. Table 1 shows mean wind velocities at four possible wind farm sites.

Table 1

Site	Mean wind velocity / m s^{-1}	Distance to grid / km	Other features
A: upland ridge	9	12	Migrating bird route; peat bog
B: coastal farmland	8	3	Near village
C: offshore shallow sea	10	15	Sandbanks
D: inland plain	6	2	Arable farmland

0 8 . 1 Calculate how many times more kinetic energy is available at site C than at site D.

[2 marks]

Answer ..... times

0 8 . 2 Use Table 1 to evaluate the four sites for a large wind farm.

[5 marks]

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0 8 . 3 Give **two** advantages of vertical axis wind turbines (VAWTs) over horizontal axis turbines (HAWTs).

[2 marks]

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0 8 . 4 Explain how direct-drive turbines improve on turbines with a gearbox.

[1 mark]

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0 9 Table 1 compares two types of satellite used to monitor the Earth.

Table 1

Feature	Low Earth Orbit (polar) satellite	Geostationary satellite
Altitude	About 800 km	36 000 km above the equator
Orbit period	About 1.5 hours	24 hours (fixed position)
Coverage	Whole Earth surveyed in about 15 days	Same third of the Earth continuously
Detail	High resolution	Lower resolution

0 9 . 1 Use Table 1 to explain why LEO satellites are used to monitor changes in ice cover and vegetation, whereas geostationary satellites are used to monitor weather.

[3 marks]

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0 9 . 2 Give **three** climate-related factors, other than ice cover, that are measured by satellite sensors.

[3 marks]

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0 9 . 3 Explain how a computer climate model can be tested for reliability before its predictions are used.

[2 marks]

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0 9 . 4 Explain why representative data are essential for climate modelling.

[2 marks]

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1 0 . 1 Explain why the development of energy storage technologies is essential if renewable energy resources are to provide most of a country's electricity.

[9 marks]

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Extra space

1 0 . 2 State what is meant by **peak shaving**.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies.

[25 marks]

OR

1 1 . 2 Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS